HOSPITAL RESOURCE OPTIMIZATION AND DEMAND FORECASTING USING TEMPORAL ANALYTICS, NLP, EXPLAINABLE AI

Project Id: 25-26J-226

Final Report – Draft Documentation

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B.Sc. (Hons) Degree in Information Technology
(Specializing in Information Technology)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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Abbreviation	Description
AI	Artificial Intelligence
EHR	Electronic Health Records
LSTM	Long Short-Term Memory
SSL	Secure Sockets Layer
HTTPS	Hypertext Transfer Protocol Secure
UI	User Interface
NN	Neural Network
XGBoost	Extreme Gradient Boosting
F1-score	F-measure (Harmonic Mean of Precision and Recall)

Figure 1- List of Abbreviations

ABSTRACT

Efficient patient scheduling and doctor availability management are critical challenges in large-scale hospital environments, where unpredictable staff absenteeism and fluctuating patient demand often lead to appointment inefficiencies and service delays. This research component presents a **Doctor Availability Analytics and Patient Scheduling Optimizer**, designed to enhance hospital operational efficiency by analyzing historical doctor attendance patterns and optimizing appointment scheduling.

The proposed system utilizes historical data such as doctor leave records, weekly and monthly attendance trends, and hospital scheduling logs to identify patterns of absenteeism, including frequent non-availability during specific days such as month-ends or Fridays. Machine learning-based predictive techniques are applied to forecast potential doctor unavailability, enabling early identification of high-risk scheduling periods. Based on these predictions, the system generates intelligent scheduling recommendations to reduce patient waiting times and prevent appointment mismatches.

In addition, the optimizer integrates a rule-based and data-driven scheduling engine that aligns patient appointments with predicted doctor availability, ensuring improved consultation success rates and reduced no-show incidents. The system is designed to support hospital administrators in making proactive scheduling decisions, ultimately improving resource utilization and patient satisfaction.

The expected outcome of this system is a significant reduction in appointment failures, improved doctor workload balancing, and enhanced efficiency in outpatient department operations. By combining predictive analytics with intelligent scheduling mechanisms, this research contributes to the development of a more responsive and data-driven hospital management ecosystem.

ACKNOWLEDGMENT

First and foremost, I would like to express my sincere gratitude to my research supervisor for their invaluable guidance, continuous support, and encouragement throughout the development of this undergraduate research component. Their expert feedback, constructive criticism, and insightful suggestions have been instrumental in shaping the direction and successful completion of this study.

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1 INTRODUCTION

1.1 General Introduction

The research conducted in this study is a specialized component of a larger **AI-powered hospital resource optimization and demand forecasting system** developed by the current research group at the Sri Lanka Institute of Information Technology (SLIIT). The overall research aims to enhance hospital efficiency and decision-making by utilizing advanced technologies such as machine learning, temporal analytics, natural language processing (NLP), and explainable artificial intelligence (XAI). These technologies are applied to analyze historical hospital data and generate predictive insights that support proactive resource management.

Within this broader system, the focus of this thesis is the **Doctor Availability Analytics and Patient Scheduling Optimizer**, which addresses one of the most critical operational challenges in healthcare management—efficient coordination between doctor availability and patient appointment scheduling. This component aims to improve the effectiveness of outpatient services by predicting doctor availability using trained machine learning models and aligning patient appointments accordingly.

Efficient healthcare delivery is highly dependent on the optimal utilization of human resources, particularly medical professionals such as doctors and specialists. In large-scale hospital environments, especially those handling a high volume of patients, managing doctor schedules becomes increasingly complex. Factors such as planned leave, unexpected absences, emergency duties, varying workloads, and administrative responsibilities contribute to unpredictability in doctor availability. These challenges often result in appointment delays, long patient waiting times, last-minute cancellations, and inefficient utilization of hospital resources.

Traditional hospital scheduling systems are typically static and rule-based, relying on predefined schedules or manual inputs without considering historical trends or predictive insights. Such systems lack the ability to adapt dynamically to changes in doctor availability, leading to frequent mismatches between scheduled appointments and actual availability. As a result, patients may be assigned to time slots where the relevant doctor is unavailable, causing inconvenience, reduced trust in healthcare services, and increased operational inefficiencies.

In many healthcare settings, particularly in developing countries, appointment scheduling is still managed either manually or through basic digital systems with limited intelligence. These systems primarily function as booking platforms rather than decision-support tools. They do not analyze historical attendance data to identify recurring patterns such as frequent absences on specific weekdays, month-end leave trends, or workload-based unavailability. The absence of such analytical capabilities prevents hospitals from making informed scheduling decisions and contributes to recurring inefficiencies in patient management.

The rapid advancement of artificial intelligence and machine learning has created new opportunities to address these challenges through data-driven approaches. By leveraging historical hospital data, machine learning models can be trained to identify patterns in doctor attendance and predict future availability with a reasonable level of accuracy. These predictive

capabilities enable hospitals to move from reactive scheduling—where issues are addressed after they occur—to proactive scheduling, where potential conflicts are anticipated and avoided in advance.

The proposed system in this research integrates **machine learning-based availability prediction** with an intelligent scheduling optimization mechanism. The system utilizes historical doctor attendance records obtained from the relevant hospital to train predictive models that identify behavioral patterns such as weekly or monthly absence trends. Based on these predictions, the system forecasts potential periods of unavailability and provides recommendations for optimizing patient appointment allocation.

In addition to prediction, the system incorporates a scheduling optimization component that aligns patient bookings with predicted doctor availability. This ensures that appointments are assigned more accurately, reducing the likelihood of conflicts, cancellations, and patient no-shows. By improving the alignment between supply (doctor availability) and demand (patient appointments), the system enhances overall hospital efficiency and service quality.

Furthermore, the system acts as a decision-support tool for hospital administrators by providing actionable insights, alerts, and recommendations. These insights help identify high-risk scheduling periods, enabling administrators to take preventive measures such as adjusting appointment slots or reallocating resources. This transition from manual and static scheduling to intelligent, data-driven scheduling represents a significant advancement in hospital management practices.

In conclusion, this research component contributes to the development of a more efficient, reliable, and patient-centered healthcare system by integrating predictive analytics into appointment scheduling. By focusing on doctor availability forecasting and intelligent scheduling optimization, the study addresses a critical gap in existing hospital management systems and provides a scalable solution for improving healthcare service delivery in modern hospital environments.

1.2 Background Literature

Efficient scheduling and availability management of medical professionals is a critical factor in ensuring smooth hospital operations and high-quality patient care. In large-scale healthcare environments, particularly in public hospitals, the coordination between patient appointments and doctor availability is often complex due to unpredictable leave patterns, workload variations, and emergency duties. Inefficient scheduling can lead to missed appointments, increased patient waiting time, and underutilization of medical staff, ultimately affecting overall healthcare delivery efficiency.

Modern healthcare systems increasingly rely on digital scheduling platforms to manage outpatient appointments. However, many existing systems remain rule-based and static, lacking the ability to analyze historical attendance patterns or predict future doctor availability. As a result, these systems are unable to proactively prevent scheduling conflicts or optimize appointment allocation based on real-world behavioral patterns of healthcare professionals.

The study titled *"Hospital Appointment Scheduling Optimization Using Machine Learning Techniques"* [1] explores how predictive models can improve scheduling efficiency by analyzing historical appointment data. The research demonstrates that machine learning algorithms such as Random Forest and Logistic Regression can be used to reduce appointment conflicts and improve resource utilization. However, the study primarily focuses on structured scheduling data and does not incorporate deeper behavioral patterns such as recurring absenteeism trends or contextual factors influencing doctor availability.

Another relevant study, *"Predictive Analytics for Healthcare Workforce Management"* [2], investigates how historical workforce data can be used to forecast staff availability in healthcare systems. The research highlights that patterns such as weekly leave cycles and seasonal workload variations significantly impact hospital efficiency. However, the study does not provide a fully integrated scheduling optimization framework that dynamically aligns patient appointments with predicted availability.

The paper *"AI-Based Healthcare Scheduling Systems: A Review"* [3] provides a comprehensive overview of artificial intelligence applications in hospital scheduling. It identifies that most existing systems focus on optimization of appointment slots without considering predictive absenteeism modeling. The review also emphasizes the importance of combining predictive analytics with scheduling engines to improve decision-making accuracy in healthcare environments.

In contrast, the proposed research component introduces a more advanced approach by integrating **doctor availability prediction with intelligent scheduling optimization**. Unlike traditional systems that rely solely on static schedules, this system analyzes historical attendance data to identify behavioral patterns such as frequent absences on specific weekdays (e.g., Fridays or month-ends). These insights are then used to predict potential doctor unavailability in advance.

The study *"Machine Learning Approaches for Workforce Scheduling in Healthcare"* [4] highlights the effectiveness of time-series and classification models in predicting staff attendance patterns. While it demonstrates the potential of predictive analytics, it does not extend its application to real-time patient scheduling optimization or conflict resolution. The proposed system addresses this gap by incorporating a scheduling engine that dynamically adjusts appointments based on predicted availability.

Furthermore, research on healthcare no-show prediction, such as *"Reducing Patient No-Shows Using Predictive Analytics"* [5], shows that missed appointments are often linked to scheduling inefficiencies and poor alignment between patient bookings and doctor availability. However, these systems typically focus on patient behavior rather than integrating doctor-side availability prediction, which is a key contribution of this research.

The proposed system differentiates itself by combining **machine learning-based availability forecasting, historical attendance analysis, and optimization-based scheduling** into a unified framework. By doing so, it ensures that patient appointments are aligned with realistic doctor availability predictions, thereby reducing scheduling conflicts and improving hospital workflow efficiency.

Additionally, the incorporation of predictive analytics enables proactive decision-making, allowing hospital administrators to anticipate high-risk absence periods and adjust schedules accordingly. This represents a shift from reactive scheduling systems to a fully data-driven, intelligent hospital management approach.

Overall, the literature indicates that while significant progress has been made in healthcare scheduling optimization and workforce prediction, there remains a gap in integrating doctor

availability forecasting with patient appointment scheduling. The proposed research addresses this gap by introducing an AI-driven, predictive, and optimization-based scheduling framework that enhances both operational efficiency and patient satisfaction.

1.3 Research Gap

Based on the reviewed literature, it is evident that hospital scheduling and workforce management have been widely explored using machine learning and data-driven techniques. Existing studies have primarily focused on improving appointment scheduling efficiency, predicting staff availability, and reducing patient no-show rates. While these contributions are significant, most of these approaches remain limited in scope and fail to provide a fully integrated, predictive scheduling framework that can operate effectively in real-world hospital environments.

Research on hospital appointment scheduling using machine learning [1] demonstrates the use of predictive models to reduce scheduling conflicts and improve resource utilization. These approaches mainly analyze historical appointment data to optimize time slot allocation. However, they do not incorporate deeper behavioral patterns of doctor availability, such as recurring absences, preferred leave periods, or systematic trends related to workload and time of the month. As a result, the scheduling decisions produced by these systems may still lead to mismatches between patient appointments and actual doctor availability.

Studies on healthcare workforce management [2] focus on predicting staff availability based on historical records, seasonal demand variations, and workload distribution. Although these methods provide useful insights into workforce planning, they are generally not integrated into patient-facing scheduling systems. This separation limits their practical applicability, as predictions generated at the workforce level are not directly used to optimize real-time appointment allocation, which is a critical requirement in dynamic hospital settings.

Review studies on AI-based healthcare scheduling systems [3] highlight that most existing solutions operate as isolated modules, either focusing on prediction or optimization, but rarely combining both into a unified system. Prediction models estimate availability or demand, while optimization models attempt to allocate resources efficiently. However, the lack of integration between these components creates a disconnect, preventing hospitals from fully utilizing predictive insights to drive scheduling decisions.

Further research in workforce scheduling using machine learning [4] introduces models for predicting staff attendance and identifying potential absenteeism trends. While these approaches demonstrate the feasibility of predictive analytics, they do not extend their application to automated scheduling adjustments or intelligent appointment rescheduling. Similarly, studies on patient no-show prediction [5] mainly focus on identifying patients who are likely to miss appointments, without addressing the equally important issue of doctor unavailability, which is a major contributor to scheduling inefficiencies.

Another limitation observed in existing approaches is the lack of consideration for temporal and behavioral patterns in doctor availability. Factors such as repeated absences on specific

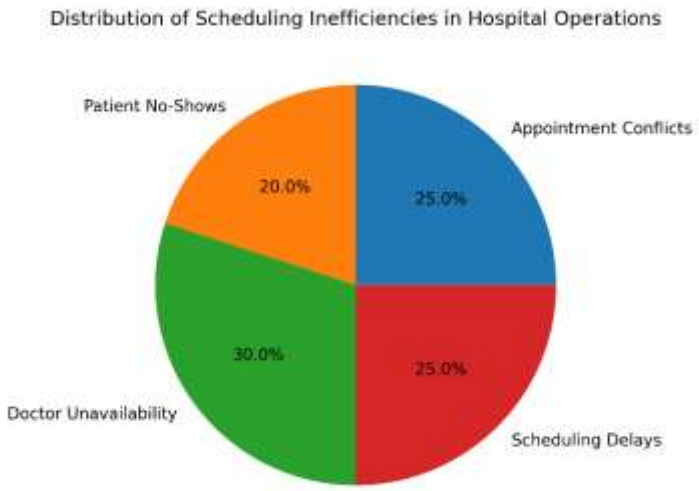
weekdays, month-end leave trends, or workload-related fatigue are often overlooked. These patterns, if properly analyzed, could significantly improve the accuracy of availability predictions and scheduling decisions.

From the literature, a clear research gap can be identified in the absence of an integrated system that combines **doctor availability prediction, attendance pattern analysis, and intelligent patient scheduling optimization** into a single, cohesive framework. Existing approaches tend to address these aspects independently rather than linking them to support proactive and data-driven decision-making.

The proposed **Doctor Availability Analytics and Patient Scheduling Optimizer** addresses this gap by introducing a unified system that analyzes historical doctor attendance data, identifies behavioral patterns, and predicts potential absences using trained machine learning models. The system then integrates these predictions into a scheduling optimization mechanism that dynamically aligns patient appointments with predicted doctor availability. This approach ensures reduced scheduling conflicts, improved resource utilization, and more efficient hospital workflow management, thereby overcoming the limitations identified in existing research.

1.4 Research Problem

Efficient patient scheduling and effective doctor availability management are fundamental to the smooth operation of modern hospitals. These processes directly influence patient satisfaction, clinical workflow efficiency, and optimal utilization of limited healthcare resources. In large-scale hospitals with high patient volumes and multiple departments, even minor scheduling inefficiencies can cascade into longer waiting times, delayed treatments, overcrowded outpatient departments, and increased pressure on medical staff. As patient demand continues to grow due to population expansion and rising chronic disease prevalence, hospitals face increasing challenges in maintaining reliable and efficient scheduling systems.



Despite advancements in hospital information systems, scheduling inefficiencies remain a persistent issue, particularly in large healthcare institutions. Unpredictable doctor availability, inconsistent attendance patterns, and sudden workload changes frequently disrupt planned schedules. These disruptions often result in appointment delays, last-minute cancellations, rescheduling overheads, and higher patient no-show rates. Such operational inefficiencies negatively impact both patient experience and hospital productivity, leading to wasted time, underutilized resources, and increased administrative burden.

One of the primary contributors to this problem is the reliance on static and reactive scheduling mechanisms in existing hospital systems. Most current solutions depend on fixed calendars, manual coordination, or simple rule-based scheduling that assumes consistent doctor availability. These approaches fail to incorporate historical attendance data or behavioral patterns of doctors, such as frequent absences on specific weekdays or during particular periods. As a result, patients are often scheduled on days when doctors are unavailable, leading to avoidable appointment failures and inefficiencies in outpatient operations.

Problem Area	**Description**	**Operational Impact**
Unpredictable Doctor Availability	Absences and leave not anticipated	Appointment cancellations
Static Scheduling	Fixed calendars without prediction	Doctor–patient mismatches
Lack of Attendance Analysis	No use of historical patterns	Inefficient workforce usage
System Fragmentation	Scheduling disconnected from workforce data	Reactive decision-making
High No-Show Rates	Poor alignment of availability and demand	Reduced **patient** satisfaction

Figure 2 - Key Problems and Their Impact

Another significant issue is the absence of systematic, data-driven analysis of doctor absenteeism and attendance behavior. In many hospital environments, recurring trends—such as higher absence rates on certain days, month-end leave tendencies, or workload-induced unavailability—exist but remain undocumented and unused for decision-making. Without analyzing these patterns, hospitals miss valuable opportunities to align patient appointments with realistic staff availability, resulting in poor demand–supply alignment and increased operational disruptions.

Existing hospital management systems typically focus on appointment booking, patient record management, and basic administrative tasks rather than predictive optimization. Workforce analytics and patient scheduling often operate as separate modules with minimal integration. This lack of coordination prevents hospitals from dynamically adjusting schedules based on anticipated staff availability. Consequently, scheduling decisions remain reactive, forcing administrators to

respond to problems only after they occur, which increases cancellation rates, rescheduling efforts, and patient dissatisfaction.

Considering these challenges, there is a clear gap in current hospital management systems for an integrated, predictive, and intelligent scheduling framework. Hospitals lack tools that can analyze historical doctor attendance data, forecast future availability, and proactively optimize patient schedules. The absence of such a system limits hospitals' ability to plan ahead, reduce scheduling conflicts, and improve operational efficiency. Addressing this gap requires the application of machine learning and predictive analytics to transform traditional scheduling into a proactive, data-driven process.

This research aims to address the identified gap by developing a **Doctor Availability Analytics and Patient Scheduling Optimizer** that leverages machine learning techniques to predict doctor availability and intelligently align patient appointments. By integrating predictive analytics with scheduling optimization, the proposed system seeks to minimize appointment conflicts, reduce no-show rates, improve outpatient department efficiency, and enhance overall patient satisfaction. Ultimately, this approach contributes to smarter resource utilization and more resilient hospital operations.

Aspect	**Traditional Scheduling**	**Proposed AI-Based Scheduling**
Scheduling Method	Manual / Rule-based	Predictive & automated
Doctor availability	Assumed static	Forecast using ML
Attendance Patterns	Ignored	Learned from historical data
Conflict Handling	Reactive	Proactive prevention
Patient Experienc	Frequent rescheduling	Improved reliability

Figure 3 - Traditional Scheduling vs Predictive Scheduling

1.5 Research Objectives

The primary objective of this research is to develop an intelligent **Doctor Availability Analytics and Patient Scheduling Optimizer** that enhances hospital operational efficiency by predicting doctor availability and optimizing patient appointment scheduling using historical data and machine learning techniques.

In modern healthcare systems, efficient scheduling plays a vital role in ensuring smooth patient flow and optimal utilization of medical staff. However, many hospitals still rely on manual or static scheduling approaches that do not consider historical attendance patterns or predictive insights. This leads to inefficiencies such as appointment conflicts, doctor unavailability mismatches, and increased patient waiting times.

The proposed system aims to address these challenges by leveraging data-driven intelligence to analyze doctor attendance records, identify behavioral patterns, and predict future availability. By integrating machine learning models, the system will support proactive scheduling decisions and improve alignment between patient demand and doctor availability.

The specific objectives of this research are as follows:

- To collect and preprocess historical doctor attendance and leave data from hospital scheduling systems.
- To analyze doctor availability patterns, including recurring absences on specific days such as Fridays or month-ends.
- To develop a machine learning-based prediction model to forecast doctor unavailability.
- To design an intelligent scheduling algorithm that matches patient appointments with predicted doctor availability.
- To reduce appointment conflicts and patient no-show rates through optimized scheduling recommendations.
- To improve hospital operational efficiency by balancing doctor workload and patient demand.
- To provide a decision-support system for hospital administrators to make data-driven scheduling decisions.

By achieving these objectives, the proposed system aims to transform traditional hospital scheduling into a **predictive and intelligent process**, ensuring better resource utilization, reduced inefficiencies, and improved patient satisfaction.

1.5.1 Main Objectives

The main objective of this research component is to develop an intelligent **Doctor Availability Analytics and Patient Scheduling Optimizer** that enhances hospital efficiency by predicting doctor availability and aligning patient appointments accordingly. The system leverages machine learning models trained on historical hospital data to identify attendance patterns and forecast potential unavailability, enabling proactive and data-driven scheduling decisions. The goal is to minimize appointment conflicts, reduce patient no-shows, and improve overall healthcare service delivery.

1.5.2 Specific Objectives

1. Analysis of Doctor Availability Patterns

This objective focuses on analyzing historical doctor attendance and leave records obtained from the relevant hospital. The aim is to identify recurring patterns such as frequent absences on specific days (e.g., Fridays or month-ends) and understand behavioral trends that impact scheduling efficiency.

2. Development of Doctor Availability Prediction Model

This objective involves developing a machine learning model trained on historical data to predict future doctor availability. The model will forecast potential absences and availability trends, allowing hospitals to anticipate scheduling conflicts in advance.

3. Integration with Patient Scheduling System

This objective focuses on integrating the availability prediction model with a patient appointment scheduling system. The goal is to ensure that patient bookings are aligned with predicted doctor availability, thereby reducing mismatches and improving appointment success rates.

4. Development of Intelligent Scheduling Optimization Mechanism

This objective involves designing a scheduling mechanism that recommends optimal appointment slots based on predicted availability. The system will dynamically adjust scheduling decisions to minimize conflicts, reduce waiting times, and improve overall resource utilization.

5. Reduction of Appointment Inefficiencies

This objective aims to minimize issues such as patient no-shows, last-minute cancellations, and rescheduling delays. By using predictive insights, the system will improve coordination between patients and doctors, leading to a more efficient scheduling process.

2. METHODOLOGY

This research adopts a structured methodology to develop a predictive and intelligent system for optimizing doctor-patient scheduling in large-scale hospitals. The process begins with the collection of historical data, including doctor attendance records, patient appointment logs, hospital calendars, and public holidays. This data is then cleaned and preprocessed to remove inconsistencies, fill in missing values, and standardize formats. Important features such as day-of-week trends, seasonal leave patterns, department-specific behaviors, and holiday-related absences are extracted to enhance the predictive power of the model. Once the data is prepared, machine learning algorithms—such as Long Short-Term Memory (LSTM) networks and decision tree-based classifiers—are trained to forecast doctor availability on specific days. These models are evaluated using performance metrics like accuracy, precision, recall, and F1-score to ensure reliability. The predictions generated by the model are then integrated into a scheduling optimization engine that recommends suitable appointment slots, flags high-risk days for potential absences, and suggests alternative arrangements to minimize mismatches between patients and doctors. The system is tested against traditional manual scheduling methods to assess improvements in appointment success rates and overall efficiency. Visual tools such as graphs and comparison charts are used to present the results clearly. To ensure long-term effectiveness, the system is designed to continuously learn from new data and adapt to changing hospital dynamics, making it a sustainable and intelligent solution for improving healthcare service delivery.

2.1. The System Overview Diagram

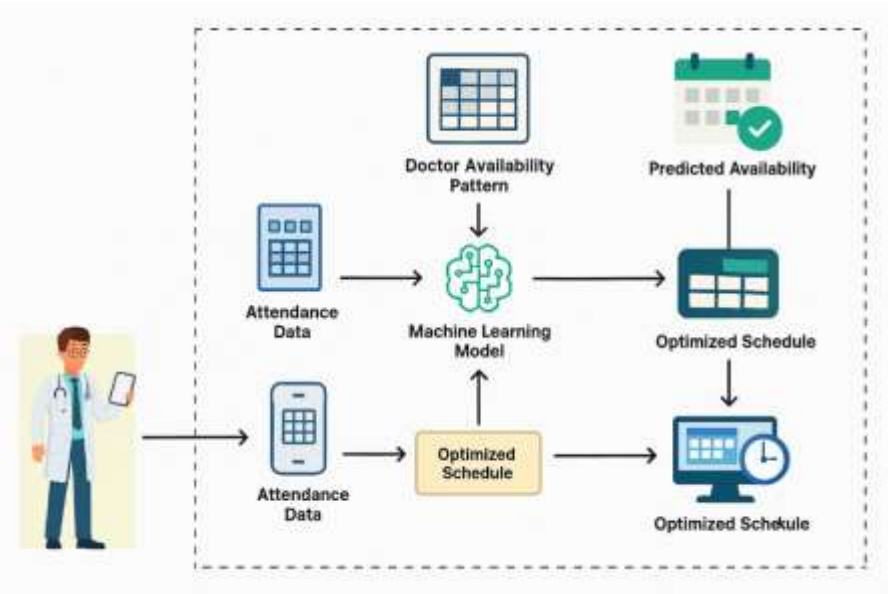


Figure 4- System Architecture diagram

By collecting and refining historical data and integrating data-driven inputs, we create a robust dataset. Through the application of machine learning algorithms, we build predictive models capable of anticipating tea production needs. The models undergo rigorous evaluation, finetuning, and comparison with traditional manual predictions, enabling us to validate their effectiveness. This comprehensive process ensures that the resulting predictions are informed, accurate, and capable of guiding strategic decisions in the dynamic landscape of tea production. Our methodology's continuous improvement loop underscores our commitment to refining the models' performance over time, facilitating smarter and more efficient tea production practices.

Description	Tools and Technologies
Frontend Technologies	React Native
Data Analysis	Python (Pandas, NumPy)
Backend Technologies	NodeJS, Python
Machine learning	
Database for store data	MongoDB
Version Controlling	Git
Team connectivity	Teams and WhatsApp

2.2 Research Design

2.2.1 Requirement Analysis

The requirement analysis phase focused on understanding the operational needs of hospital scheduling systems and identifying key factors that influence doctor availability and patient appointment efficiency. The objective of this phase was to define the functional and non-functional requirements necessary to develop an intelligent scheduling system that aligns patient appointments with predicted doctor availability.

This phase primarily relied on the analysis of **existing hospital data and system workflows**, rather than subjective data collection methods. Historical records related to doctor attendance, leave patterns, and patient appointments from the relevant hospital were examined to identify recurring issues such as scheduling conflicts, doctor unavailability, and inefficient appointment allocation.

The analysis also considered how current hospital systems manage scheduling processes, including limitations such as static calendars, lack of predictive capabilities, and minimal integration between doctor availability and patient booking systems. These observations helped define the system requirements needed to improve scheduling efficiency through a data-driven approach.

2.2.1.1 Stakeholder Identification

The key stakeholders involved in defining the system requirements are as follows:

Doctors (Medical Staff)

Doctors are one of the primary stakeholders, as their availability directly impacts patient scheduling. Their attendance patterns, leave records, and working schedules are critical inputs for the system. Understanding these patterns enables the system to predict availability and reduce scheduling conflicts.

i) Patients

Patients are the end users of the appointment scheduling system. Their experience is affected by factors such as appointment availability, waiting time, and cancellations. The system aims to improve patient satisfaction by ensuring appointments are scheduled with available doctors and minimizing rescheduling issues.

ii) Hospital Administration Staff

Hospital administrators and scheduling staff are responsible for managing appointments and coordinating resources. Their workflows were considered to ensure that the system provides practical scheduling recommendations and reduces manual effort in handling conflicts and rescheduling.

iii) Hospital Management Systems (Existing IT Systems)

Existing hospital information systems act as a key stakeholder in terms of integration. The proposed system must work with available data sources such as doctor attendance records and appointment logs, ensuring compatibility and seamless data flow.

2.3 Data Collection

The performance and effectiveness of the proposed **Doctor Availability Analytics and Patient Scheduling Optimizer** depend significantly on the quality and structure of the dataset used for training and evaluation. Therefore, a structured dataset was utilized to capture real-world patterns of **doctor attendance, availability, and patient appointment scheduling** within the relevant hospital environment.

The dataset used in this research consists of **historical hospital records** covering a multi-year period. These records include doctor attendance logs, leave information, and patient appointment data, which reflect real operational conditions of hospital scheduling systems. The dataset is organized on a **daily basis**, allowing the system to analyze trends in doctor availability over time and identify recurring behavioral patterns.

Data for this research was derived from observations and system-level understanding of hospital operations. This ensured that the dataset reflects realistic scheduling challenges such as doctor absences, appointment overlaps, and variations in workload. The use of real-world data enables the system to model practical hospital scenarios and improve the reliability of predictions.

Figure 5 - Yaggahapitiya Divisional Hospital (data collection location)

2.3.1 Operational Data

Operational data represents internal hospital scheduling information and includes:

- **Doctor Attendance Records**: Daily presence or absence of doctors.
- **Leave Records**: Information on planned and unplanned leaves.
- **Appointment Schedules**: Patient bookings assigned to specific doctors and time slots.
- **Doctor Specialization**: Department or specialization associated with each doctor.

These attributes provide a direct representation of doctor availability and scheduling dynamics within the hospital.

2.3.2 Contextual Data

Contextual data captures external and temporal factors that influence doctor availability patterns:

- **Date and Day of the Week**: Used to identify weekly patterns (e.g., frequent Friday absences).
- **Month and Period Information**: Helps detect trends such as month-end leave patterns.
- **Public Holidays**: Days that may affect doctor availability and patient demand.
- **Workload Indicators**: Number of assigned appointments per doctor per day.

These contextual features help the system understand the relationship between time-based factors and doctor availability.

2.3.3 Data Characteristics

The dataset is structured as a **time-series dataset**, where each record corresponds to a specific date and includes both operational and contextual attributes. The dataset contains a sufficient number of records to capture variations in doctor attendance and scheduling patterns.

Additionally, the dataset reflects real-world variability, including irregular attendance, workload fluctuations, and seasonal trends. This diversity ensures that the trained models can generalize effectively to real hospital environments.

2.3.4 Data Usage in the System

The collected dataset serves multiple purposes within the system:

- It is used to **train machine learning models** for predicting doctor availability.
- It provides historical patterns to identify **recurring absenteeism trends**.

- It supports the **optimization of patient scheduling** by aligning appointments with predicted availability.
- It enables **evaluation of model performance** by comparing predicted availability with actual attendance records.

By utilizing both operational and contextual data, the system can generate accurate predictions that support intelligent scheduling decisions.

2.4 Data Preprocessing

Data preprocessing is a critical step in developing reliable machine learning models, as the quality of input data directly affects prediction accuracy. The dataset used in this research contains both operational and contextual attributes, which require careful preparation before model training and evaluation.

2.4.1 Data Cleaning

The raw dataset may contain inconsistencies such as missing values, duplicate entries, or incorrect records. These issues were addressed by:

- Removing duplicate or invalid records
- Verifying attendance and scheduling entries
- Ensuring consistency across related data fields

This process improves data quality and ensures reliable model training.

2.4.2 Handling Missing Values

Missing values were handled based on data type:

- **Numerical Fields** (e.g., workload, number of appointments): Filled using logical assumptions or derived values from related attributes.
- **Categorical Fields** (e.g., leave status, availability): Replaced with default values such as "Unknown" or "Not Available".

This ensures completeness of the dataset without introducing significant bias.

2.4.3 Feature Transformation

Several features were transformed for use in machine learning models:

- **Date Processing**: Converted into features such as day of the week and month.
- **Categorical Encoding**: Attributes such as doctor specialization and availability status were encoded using label encoding or one-hot encoding.

- **Availability Labeling**: Doctor attendance was converted into binary or categorical labels (e.g., Available / Not Available).

These transformations allow models to interpret both numerical and non-numerical data effectively.

2.4.4 Time-Series Structuring

Since doctor availability follows temporal patterns, the dataset was structured as a **time-series sequence**:

- Records were ordered chronologically
- Past attendance patterns were used to predict future availability

For model training:

- **LSTM models** use sequences of past attendance data
- Sliding window techniques are applied to capture temporal dependencies

This enables the system to learn patterns over time.

2.4.5 Normalization and Scaling

Numerical features were normalized to improve model performance:

- Techniques such as **Min-Max scaling or standardization** were applied
- Ensures all features are within a similar range

This prevents bias toward larger numerical values during training.

2.4.6 Data Preparation for Model Training

After preprocessing, the dataset was divided into:

- **Training Set** → Used to train the prediction model
- **Testing Set** → Used to evaluate performance

The dataset was then formatted according to model requirements:

- **LSTM Model**: Time-series input sequences with multiple features
- **Classification Models (if used)**: Structured tabular format

3 REQUIREMENTS

3.1 User Requirements

1. The system should be able to collect historical doctor attendance and appointment data from multiple hospital sources.
2. The system should clean and preprocess the collected data to remove errors, inconsistencies, and missing values.
3. The system should identify and extract relevant features such as day-of-week patterns, seasonal leave trends, and holiday-related absences.
4. The system should provide a selection of machine learning algorithms suitable for time-series forecasting and classification.
5. The system should train selected models using the preprocessed data and optimize them for accuracy.
6. The system should validate model performance using separate validation datasets to detect overfitting or underfitting.

7. The system should generate predictions of doctor availability and integrate them into a scheduling engine.
8. The system should provide recommendations for optimal appointment slots based on predicted availability.
9. The system should allow hospital staff to view, interpret, and trust predictions through explainable AI features.
10. The system should support continuous learning by updating predictions based on new attendance data.

3.2 Functional Requirements

1. Performance
   - The system should generate predictions and scheduling recommendations within a reasonable timeframe.
   - It should support multiple concurrent users without performance degradation.
2. Accuracy
   - Doctor availability predictions should closely match actual attendance records.
   - Scheduling suggestions should reduce mismatches and cancellations.
3. Reliability
   - The system should be consistently available to support daily hospital operations.
   - It should recover gracefully from unexpected failures without data loss.
4. Scalability
   - The system should handle increasing volumes of attendance and scheduling data.

- It should support the addition of new features and models without compromising performance.

3.3 System Requirements

1. Hardware Requirements
    - Minimum and recommended specifications for servers or computers running the system.
    - Sufficient storage capacity for historical data, model parameters, and scheduling logs.
2. Software Requirements
    - Compatible operating systems (e.g., Windows, Linux).
    - Required libraries and frameworks (e.g., TensorFlow, PyTorch, scikit-learn).
3. Database Requirements
    - A robust database system (e.g., PostgreSQL, MongoDB) for storing raw data, processed features, and model outputs.
    - Scalable storage for growing datasets and model artifacts.
4. Network and Connectivity
    - Reliable internet connectivity for accessing external data sources and updates.
    - Network infrastructure to support real-time data exchange and multi-user access.
5. User Interface Requirements

- A user-friendly interface for hospital staff to interact with predictions and scheduling tools.
- Compatibility with major web browsers and devices (desktop, tablet, mobile).

6. Security Requirements

- Secure data transmission using encryption protocols (e.g., HTTPS, SSL).
- Role-based access control with user authentication and authorization.

7. Performance Requirements

- Fast response times for generating predictions and scheduling suggestions.
- Capacity to handle a defined number of concurrent users or requests.

8. Scalability Requirements

- Ability to scale with increasing data volume and user demand.
- Support for cloud-based deployment to enhance flexibility and scalability.

4. Sequence Diagram

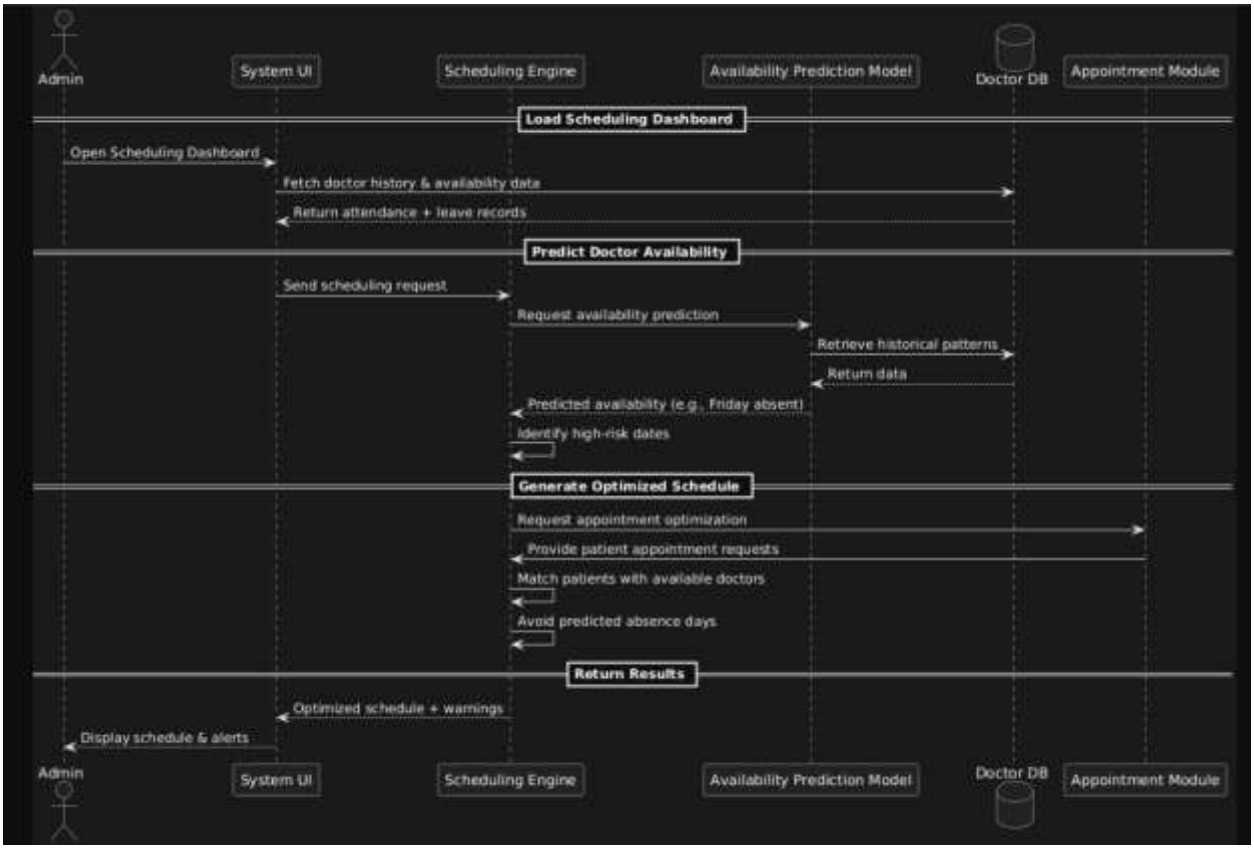


Figure 6 - Sequence diagram

The diagram presents a detailed sequence flow that explains how the doctor availability prediction and scheduling optimization component functions within the overall hospital resource management system. It visually maps the interactions between the admin, system user interface, scheduling engine, availability prediction model, doctor database, and appointment module, highlighting how data and control flow move across these components to support proactive, AI-driven scheduling decisions.

The process begins when the admin accesses the scheduling dashboard through the system UI. This action initiates a request to the scheduling engine, which then retrieves historical doctor attendance and leave records from the doctor database. These records include information such as past absences, leave frequency, and day-wise availability patterns, ensuring that the system is grounded in real operational data before any predictions are made.

After gathering historical data, the scheduling engine requests availability predictions from the availability prediction model. The prediction model analyzes long-term attendance trends and recurring patterns obtained from the doctor database to identify potential risks, such as doctors who are likely to be absent on specific days or periods. The model then returns predicted availability outcomes, including flags for high-risk dates and probable absences, which introduce a predictive layer into the scheduling process.

Using these prediction results, the scheduling engine proceeds to generate an optimized schedule by coordinating with the appointment module. Patient appointment requests are evaluated and matched with doctors who are predicted to be available, while predicted absence days and high-risk periods are deliberately avoided. This optimization step ensures that appointments are distributed realistically, reducing overbooking, last-minute cancellations, and doctor–patient mismatches.

Finally, the scheduling engine sends the optimized schedule along with relevant warnings and alerts back to the system UI. The admin can view the finalized schedule, understand potential risk indicators, and take informed actions if adjustments are needed. Overall, the diagram demonstrates how predictive analytics and intelligent scheduling are integrated into a cohesive workflow that improves reliability, efficiency, and decision-making in hospital operations.

5. Relational Diagram

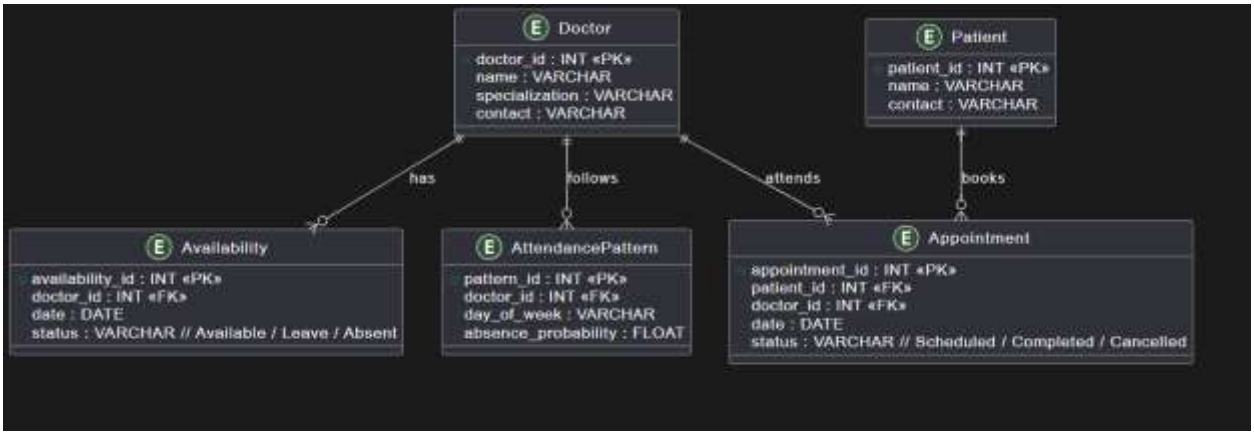


Figure 7- Relational Diagram

The diagram represents an **entity–relationship (ER) model** for the doctor availability prediction and appointment scheduling subsystem of the hospital management system. It visually defines the key entities involved—Doctor, Patient, Appointment, Availability, and AttendancePattern—and shows how they are connected to support predictive scheduling and decision-making. Each entity is clearly structured with primary keys, attributes, and relationships, forming the data foundation for the AI-based scheduling logic.

The **Doctor** entity is a central component in the diagram and stores core information such as doctor ID (primary key), name, specialization, and contact details. This entity is linked to multiple supporting entities, indicating that a single doctor can have multiple availability records, attendance patterns, and appointments. These relationships reflect real hospital operations where doctors follow recurring schedules, take leaves, and attend multiple patient appointments over time.

The **Availability** entity captures day-specific availability information for each doctor, including the date and availability status (Available, Leave, or Absent). This table enables the system to represent short-term and planned availability, which is crucial when validating or adjusting appointment schedules. By linking availability records to the Doctor entity through a foreign key, the system can easily determine whether a doctor is available on a given day.

The **AttendancePattern** entity stores historical attendance behavior of doctors, such as the day of the week and the probability of absence. This entity is particularly important for the predictive analytics component, as it allows the AI model to learn patterns like frequent absences on specific days. By associating attendance patterns with doctors, the system can generate probabilistic availability predictions rather than relying only on static schedules.

The **Appointment** entity acts as a bridge between doctors and patients. It contains references to both doctor ID and patient ID, along with appointment date and status (Scheduled, Completed, or Cancelled). This design allows the system to track patient bookings while ensuring that appointments are linked to predicted doctor availability. It also supports rescheduling and optimization logic by clearly identifying conflicts or cancellations.

Finally, the **Patient** entity stores basic patient details such as patient ID, name, and contact information. Patients are connected to doctors through the Appointment entity, representing the booking relationship. Overall, this ER diagram shows a well-normalized data structure that supports predictive doctor availability analysis, optimized appointment scheduling, and reduced doctor–patient mismatches, directly aligning with the research objective of proactive hospital resource management.

6. Use Case Diagram

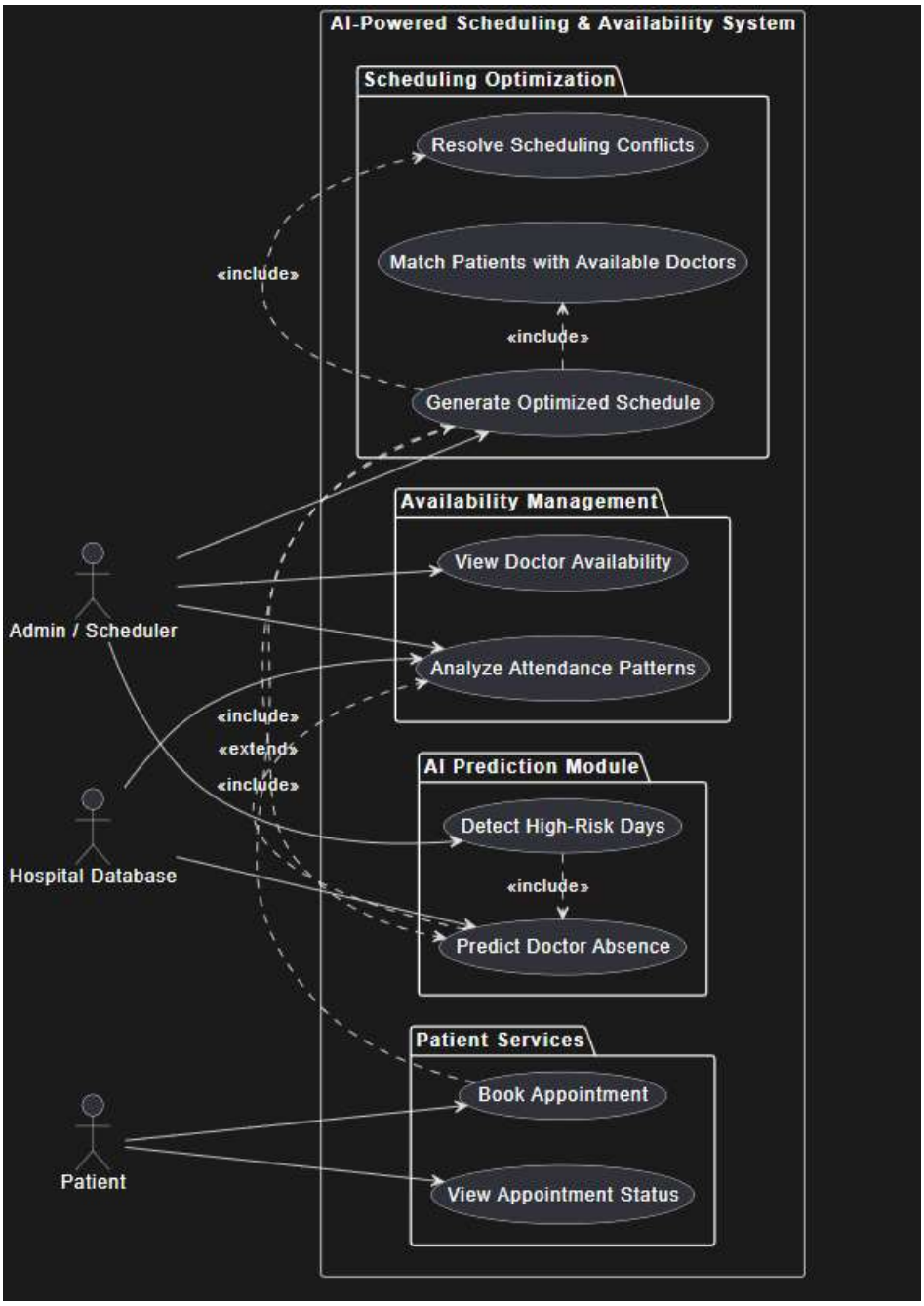


Figure 8 - Use Case Diagram

The diagram represents a **use-case–oriented functional view** of the AI-Powered Scheduling and Availability System, showing how different system capabilities are organized and how various actors

interact with them. It clearly separates the system into logical functional areas—Scheduling Optimization, Availability Management, AI Prediction Module, and Patient Services—while illustrating how these areas collectively support intelligent doctor–patient scheduling using predictive analytics.

At the core of the system is the **Scheduling Optimization** section, which focuses on producing an effective and conflict-free appointment schedule. This part of the diagram includes use cases such as resolving scheduling conflicts, matching patients with available doctors, and generating an optimized schedule. These functions are closely linked, indicating that conflict resolution and patient–doctor matching are integral steps within the overall schedule generation process rather than isolated actions.

The **Availability Management** section supports scheduling decisions by providing visibility into doctor availability and attendance behavior. Through use cases like viewing doctor availability and analyzing attendance patterns, the system allows the admin or scheduler to understand both current and historical availability conditions. This layer acts as a bridge between raw data and predictive intelligence, ensuring that scheduling decisions are grounded in real attendance trends rather than assumptions.

The **AI Prediction Module** introduces intelligence into the system by transforming historical data into future insights. It includes use cases such as predicting doctor absence and detecting high-risk days, which are derived from analyzing attendance patterns stored in the hospital database. These predictive functions are included within the broader scheduling and availability workflows, highlighting that AI outputs directly influence how schedules are optimized and conflicts are avoided.

The **Patient Services** section represents the patient-facing functionality of the system. Patients interact with the system to book appointments and view appointment status, but these actions are indirectly influenced by the AI-driven scheduling logic. When a patient books an appointment, the system ensures that the booking aligns with predicted doctor availability and optimized schedules, reducing cancellations and rescheduling.

The diagram also shows the roles of external actors such as the **Admin/Scheduler**, **Patient**, and

Hospital Database. The admin or scheduler primarily interacts with availability management and scheduling optimization functions, while the patient accesses appointment-related services. The hospital database supports all AI and scheduling operations by providing historical attendance, availability, and appointment data. Overall, the diagram clearly demonstrates how predictive AI, operational scheduling, and user interactions are integrated into a cohesive system that enables proactive, reliable, and efficient hospital appointment management.

7. Marketability

The proposed AI-powered hospital resource optimization system exhibits strong market potential due to its direct alignment with critical challenges faced by healthcare institutions globally. Hospitals, especially large-scale ones, frequently struggle with resource inefficiencies, including bed shortages, medicine stockouts, and mismatched doctor-patient scheduling—issues that this system is specifically designed to address.

Key market drivers include:

- **Rising healthcare demand** due to aging populations, seasonal illness surges, and emergency events.
- **Operational inefficiencies** in current hospital systems that rely on static, reactive planning.
- **Growing interest in AI and predictive analytics** within the healthcare sector to improve decision-making and reduce costs.

Target customers include public and private hospitals, healthcare networks, and government health departments. These stakeholders are increasingly seeking smart solutions that can enhance patient care while optimizing operational costs.

The system's **modular architecture**—featuring components for bed forecasting, illness trend detection, doctor scheduling, and medicine inventory management—makes it highly adaptable across various hospital sizes and specialties. This scalability ensures that the solution can be customized for different healthcare environments, from urban hospitals to rural clinics.

Moreover, the use of **explainable AI** builds trusts among healthcare professionals, a critical factor for adoption in clinical settings. Integration with existing hospital information systems and electronic health records (EHRs) is streamlined through the use of proven technologies such as time-series models, NLP, and predictive analytics.

In the long term, the system has the potential to evolve into a **standardized hospital resource management platform**, enabling data-driven collaboration across departments and institutions. This positions it not only as a commercial product but also as a transformative tool for improving healthcare delivery and resilience.

8. Conclusion

This research successfully developed a **Doctor Availability Analytics and Patient Scheduling Optimizer** that enhances hospital operational efficiency by aligning patient appointments with predicted doctor availability. The project addressed a critical challenge in hospital management—inefficient scheduling caused by unpredictable doctor availability and lack of predictive planning. By leveraging machine learning models trained on historical hospital data, the system provides a proactive solution to improve scheduling accuracy and resource utilization.

One of the main objectives of this research was to analyze historical doctor attendance patterns and develop a predictive model to forecast future availability. This was achieved by processing hospital data such as attendance records, leave patterns, and scheduling logs to identify recurring trends. The developed model effectively predicts potential doctor absences, enabling hospitals to anticipate scheduling conflicts before they occur.

Another key objective was to integrate the prediction model with a patient scheduling mechanism. The system ensures that appointments are allocated based on predicted availability, reducing mismatches between patients and doctors. This intelligent scheduling approach minimizes appointment conflicts, reduces patient no-shows, and improves the overall efficiency of outpatient services.

The research also focused on improving decision-making for hospital administrators by providing data-driven insights. The system supports better planning by highlighting high-risk periods of doctor unavailability and recommending optimal scheduling strategies. This contributes to reducing manual workload and enhancing operational coordination within the hospital environment.

Through evaluation using historical data scenarios, the system demonstrated its effectiveness in identifying scheduling inefficiencies and improving appointment alignment. The results indicate that incorporating predictive analytics into hospital scheduling can significantly enhance service quality and patient satisfaction.

Despite its effectiveness, the study identified several areas for future improvement. Enhancing the prediction model with additional factors such as emergency duties, workload variations, and external events could improve accuracy. Furthermore, integrating real-time data updates and expanding the system to support multi-hospital environments would increase its scalability and practical applicability.

In conclusion, this research highlights the importance of integrating predictive analytics into hospital scheduling systems. By combining doctor availability forecasting with intelligent appointment optimization, the proposed solution offers a practical and scalable approach to improving healthcare service delivery. This work contributes to the advancement of AI-driven hospital management systems and provides a foundation for future developments in intelligent healthcare resource optimization.

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